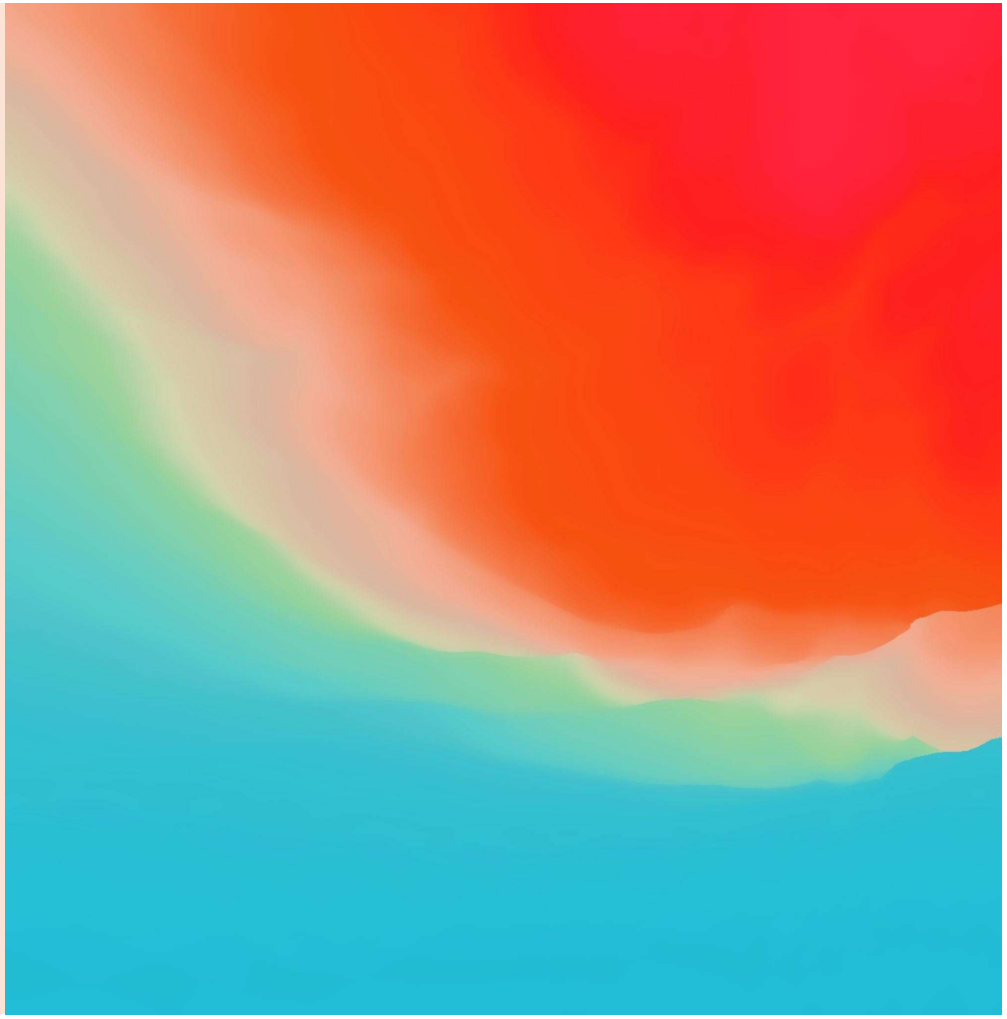
Week 5 (Extra)

Nonlinear programming



Eigenvalues

Recall how to find eigenvalues of a square matrix A:

Obtain the roots for $0 = \det(A - \lambda I_n)$ or $0 = \det(\lambda I_n - A)$.

Example. Find the eigenvalues for $\begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}$.

$$0 = \det\begin{pmatrix} \lambda - a & 0 & 0 \\ 0 & \lambda - b & 0 \\ 0 & 0 & \lambda - c \end{pmatrix} = (\lambda - a)(\lambda - b)(\lambda - c)$$

So the eigenvalues are a, b and c.

Useful results

Theorem. Suppose A is a real symmetric matrix.

Then all eigenvalues of A are real numbers.

Theorem. Suppose $D = \mathrm{diag}(\lambda_1, \lambda_2, ..., \lambda_n)$ is a diagonal maxtrix.

Then $\lambda_1, \lambda_2, ..., \lambda_n$ are all eigenvalues of D.

Example

Example. Find the eigenvalues for $\begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$.

$$0 = \det\begin{pmatrix} 2-\lambda & -1 & 0 \\ -1 & 2-\lambda & -1 \\ 0 & -1 & 2-\lambda \end{pmatrix} = (2-\lambda)\begin{vmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{vmatrix} + \begin{vmatrix} -1 & -1 \\ 0 & 2-\lambda \end{vmatrix}$$

$$= (2-\lambda)((2-\lambda)^2-1) - (2-\lambda) = (2-\lambda)((2-\lambda)^2-2) \Rightarrow \lambda = 2, 2\pm\sqrt{2}$$

Uncontrained nonlinear programming

Suppose $f(\mathbf{x}) : \mathbb{R}^n \to \mathbb{R}$ is a function of C^2 class at point $\mathbf{x} \in \mathbb{R}^n$.

We introduce the n by n Hessian matrix H of $f(\mathbf{x})$, defined as

$$H(\mathbf{x}) = \begin{pmatrix} f_{x_1x_1}(\mathbf{x}) & f_{x_1x_2}(\mathbf{x}) & \cdots & f_{x_1x_n}(\mathbf{x}) \\ f_{x_2x_1}(\mathbf{x}) & f_{x_2x_2}(\mathbf{x}) & \cdots & f_{x_2x_n}(\mathbf{x}) \\ \vdots & \vdots & \ddots & \vdots \\ f_{x_nx_1}(\mathbf{x}) & f_{x_nx_2}(\mathbf{x}) & \cdots & f_{x_nx_n}(\mathbf{x}) \end{pmatrix}.$$

The Hessian matrix represents the second derivative of a function with more than one variable, sometimes some authors write $\nabla^2 f(\mathbf{x})$ for $H(\mathbf{x})$.

The second partial derivatives test

To determine the local max or local min of $f(\mathbf{x})$.

Let $\mathbf{x}^* \in \mathbb{R}^n$ be a point such that the following conditions are satisfied:

Necessary conditions

1. $\nabla f(\mathbf{x}^*) = 0$

Sufficient conditions

2. Form the Hessian matrix, $H(\mathbf{x}^*)$. Then $\mathbf{x}^* \in \mathbb{R}^n$ is
   (a) a local minimum if $H(\mathbf{x}^*)$ is positive definite
   (b) a local maximum if $H(\mathbf{x}^*)$ is negative definite
   (c) a saddle point if $H(\mathbf{x}^*)$ has both positive and negative eigenvalues
   (d) inconclusive if it is not listed above.

Example 1

Minimise $f(x,y,z) = x^2 + x(1-y) + y^2 - yz + z^2 + z\,.$

Solution:

$$\nabla f(x,y,z) = 0 \Rightarrow \begin{pmatrix} f_x \\ f_y \\ f_z \end{pmatrix} = 0 \Rightarrow \begin{cases} 2x + 1 - y = 0 \\ -x + 2y - z = 0 \Rightarrow x = y = z = -1. \\ -y + 2z + 1 = 0 \end{cases}$$

So, we only have one point to verify here, i.e., $(-1,-1,-1)\,.$

$$H(x,y,z)=\begin{pmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{pmatrix}=\begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}.$$

$$H(-1,-1,-1)=\begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}.$$

Now determine if $H(-1,-1,-1)$ is positive definite.

$$\left|H(-1,-1,-1)-\lambda I_3\right|=(2-\lambda)(\lambda^2-4\lambda+2)=0\Rightarrow\lambda=2,2\pm\sqrt{2}>0\,.$$

Since the eigenvalues are all positive, it follows that $H(-1,-1,-1)$ is positive definite.

So, the point $(-1,-1,-1)$ is a local minimum point.

Example 2

Maximise $f(x,y,z) = -x^2 - 5y^2 - z^2 + 4xy + 2y - 1\,.$

Solution. $\nabla f = 0 \Rightarrow \begin{cases} -2x + 4y = 0 \\ -10y + 4x + 2 = 0 \\ -2z = 0 \end{cases} \Rightarrow (x,y,z) = (2,1,0)\,.$

$H(x,y,z) = \begin{pmatrix} -2 & 4 & 0 \\ 4 & -10 & 0 \\ 0 & 0 & -2 \end{pmatrix}$. Eigenvalues $= -6 \pm 4\sqrt{2}, -2 \; < 0.$

So $(2,1,0)$ is a local maximum. $f_{\max}(2,1,0) = 0\,.$

Alternative: Write $f(x,y,z) = -(x-2y)^2 - (y-1)^2 - z^2\,.$

D test Proof (from Hessian matrix)

Consider a Hessian matrix

$$H(x,y) = \begin{pmatrix} f_{xx}(x,y) & f_{xy}(x,y) \\ f_{yx}(x,y) & f_{yy}(x,y) \end{pmatrix}.$$

We find the corresponding eigenvalues of the matrix, i.e.,

$$0 = \begin{vmatrix} f_{xx}(a,b) - \lambda & f_{xy}(a,b) \\ f_{yx}(a,b) & f_{yy}(a,b) - \lambda \end{vmatrix} = (f_{xx}(a,b) - \lambda)(f_{yy}(a,b) - \lambda) - f_{xy}^2(a,b)$$

$$\Rightarrow 0 = \lambda^2 - (f_{xx}(a,b) + f_{yy}(a,b))\lambda + (f_{xx}(a,b)f_{yy}(a,b) - f_{xy}^2(a,b)),$$

assuming that $f_{xy}(a,b) = f_{yx}(a,b)$.

Note that the roots are eigenvalues. Recall that for $x^2 + Bx + C = 0$, $B^2 - 4C \geq 0$ iff all roots are real; C is the product of roots and $-B$ is the sum of the roots.

Let $D(a,b) = f_{xx}(a,b)f_{yy}(a,b) - f_{xy}^2(a,b)$ is the product of roots.

First, we prove that eigenvalues are all real:

$$\begin{aligned}
&(f_{xx}(a,b) + f_{yy}(a,b))^2 - 4(f_{xx}(a,b)f_{yy}(a,b) - f_{xy}^2(a,b)) \\
&= f_{xx}^2(a,b) + f_{yy}^2(a,b) - 2f_{xx}(a,b)f_{yy}(a,b) + 4f_{xy}^2(a,b) \\
&= (f_{xx}(a,b) - f_{yy}(a,b))^2 + f_{xy}^2(a,b) \geq 0
\end{aligned}$$

Case 1. $D(a,b) = 0$. Then either one of the eigenvalues is zero. From the second derivative test (condition 2d), this gives an inconclusive result.

Case 2. $D(a,b) = f_{xx}(a,b)f_{yy}(a,b) - f_{xy}^2(a,b) < 0$ (product of roots is negative). Then sure the eigenvalues are of different sign, then from the second derivative test (condition 2c),

(a,b) is a saddle point.

Case 3. $D(a,b) = f_{xx}(a,b)f_{yy}(a,b) - f_{xy}^2(a,b) > 0$ (eigenvalues of the same sign then the product of roots is positive). Then from the second derivative test (condition 2a and 2b), (a,b) is either a local minimum or local maximum.

Case 3(a): Suppose $f_{xx}(a,b) > 0$. Then $f_{yy}(a,b) > 0$ (otherwise,

$f_{xx}(a,b)f_{yy}(a,b) \leq 0$ and $D(a,b) = f_{xx}(a,b)f_{yy}(a,b) - f_{xy}^2(a,b) \leq 0$,

a contradiction). Since $f_{xx}(a,b), f_{yy}(a,b) > 0$, it follows that

$-B = f_{xx}(a,b) + f_{yy}(a,b) > 0$ (i.e., sum of the roots is positive).

So, the eigenvalues are positive. It follows that (a,b) is a local minimum.

The argument for other cases is similar.

Constrained nonlinear programming : Karush-Kuhn-Tucker method

Suppose that $f, g_i, h_j : \mathbb{R}^n \to \mathbb{R}$ are functions of C^2 class at point $\mathbf{x}^* \in \mathbb{R}^n$.

Consider the following nonlinear optimisation problem in standard form

Minimise/maximise	objective function $f(\mathbf{x})$
subject to	equality constraints $g_i(\mathbf{x}) = 0$,
	and inequality constraints $h_j(\mathbf{x}) \leq 0$

where $\mathbf{x} \in \mathbb{R}^n$, $i = 1,\ldots,m, j = 1,\ldots,l$.

To determine the max or min of $f(\mathbf{x})$ in the domain satisfying the constraints:

Let $\mathbf{x}^* \in \mathbb{R}^n$ be an optimal solution, then there exist constants

$\lambda_1,...,\lambda_m$, (called the Lagrange multipliers) and

$\mu_1,...,\mu_l$, (called the KKT multipliers),

such that the following conditions are satisfied:

Necessary conditions

1. Stationarity:

For minimising $f(\mathbf{x})$: $\nabla\left(f(\mathbf{x}^*)+\sum_{i=1}^{m}\lambda_i g_i(\mathbf{x}^*)+\sum_{j=1}^{l}\mu_i h_j(\mathbf{x}^*)\right)=0$

For maximising $f(\mathbf{x})$: $\nabla\left(f(\mathbf{x}^*)-\sum_{i=1}^{m}\lambda_i g_i(\mathbf{x}^*)-\sum_{j=1}^{l}\mu_i h_j(\mathbf{x}^*)\right)=0$

2. Primal feasibility: (a) $g_i(\mathbf{x}^*) = 0, \;\; i = 1,\ldots,m$

(b) $h_j(\mathbf{x}^*) \leq 0, \;\; j = 1,\ldots,l$

3. Dual feasibility: $\mu_j \geq 0, \;\; j = 1,\ldots,l$

4. Complementary slackness: $\mu_j h_j(\mathbf{x}^*) = 0, \;\; j = 1,\ldots,l$

Remark: When $m = 0$, i.e., when there are no inequality constraints, then the KKT turn into Lagrange multiplier.

Example 3

Using KKT method:

Minimise $\quad f(x,y) = x^2 - xy + y^2 - 3x$

Subject to $\quad g(x,y) = x^2 + y^2 - 9 \leq 0\,.$

Solution. Stationarity for minimising

$$\nabla\Big(f + \mu g\Big) = 0 \Rightarrow \begin{pmatrix} 2x - y - 3 + 2\mu x \\ 2y - x + 2\mu y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Stationarity	$2(1+\mu)x = y+3$	(1)
Stationarity	$2(1+\mu)y = x$	(2)
Primal feasibility	$x^2+y^2 \le 9$	(3)
Dual feasibility	$\mu \ge 0$	(4)
Complementary slackness	$\mu(x^2+y^2-9)=0$	(5)

$(1)\times y = (2)\times x:\ y^2+3y = x^2$ sub into (5):

$\mu(2y^2+3y-9)=0 \Rightarrow \mu(2y-3)(y+3)=0 \Rightarrow \mu=0,$ or $y=-3$, or $y=\frac{3}{2}$.

Case 1. If $\mu=0$. Then from (1) and (2) we have $\begin{cases} 2x = y+3 \\ 2y = x \end{cases}$.

So, $(x,y)=(2,1)$.

Case 2. If $\mu \neq 0$ and when $y = -3$ or $\frac{3}{2}$. Then from (5), $x = 0$ or $\pm\frac{3\sqrt{3}}{2}$.

Check that:

(x, y)	$\mu = \frac{x}{2y} - 1$ (from (2))	Conclusion
$\left(\frac{3\sqrt{3}}{2}, \frac{3}{2}\right)$	$\frac{-2+\sqrt{3}}{2} < 0$, violates (4)	rejected
$\left(-\frac{3\sqrt{3}}{2}, \frac{3}{2}\right)$	$-\frac{2+\sqrt{3}}{2} < 0$, violates (4)	rejected
$(0, -3)$	$-1 < 0$, violates (4)	rejected

Case 3. Obviously, we cannot have $y = -3$ and $y = \frac{3}{2}$.

One can check these cases $(y, \mu) = (-3, 0)$ or $(\frac{3}{2}, 0)$.

From (2), we have $x = -6, 3$.

Check that:

(x, y)	$x^2 + y^2 \leq 9$ (3)	Conclusion
$\left(-6, 3\right)$	violates (3)	rejected
$\left(3, \frac{3}{2}\right)$	violates (3)	rejected

Choose $(x, y) = (0, 0)$. The point satisfies $g(x, y) \leq 0$.

We see that $f(0, 0) = 0 > -3 = f(2, 1)$, which verify

that $f_{\min} = -3$ at $(2, 1)$.

Example 4 (same as Example 3 but this is maximising)

Using KKT method:

Maximise $\quad f(x,y) = x^2 - xy + y^2 - 3x$

Subject to $\quad g(x,y) = x^2 + y^2 - 9 \leq 0\,.$

Solution. Stationarity for maximising

$$\nabla\big(f - \mu g\big) = 0 \Rightarrow \begin{pmatrix} 2x - y - 3 - 2\mu x \\ 2y - x - 2\mu y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

A different sign when maximising!!!

Stationarity	$2(1-\mu)x = y+3$	(1)
Stationarity	$2(1-\mu)y = x$	(2)
Primal feasibility	$x^2+y^2 \le 9$	(3)
Dual feasibility	$\mu \ge 0$	(4)
Complementary slackness	$\mu(x^2+y^2-9)=0$	(5)

$(1)\times y = (2)\times x:\ y^2+3y = x^2$ sub into (5):

$\mu(2y^2+3y-9)=0 \Rightarrow \mu(2y-3)(y+3)=0 \Rightarrow \mu=0,$ or $y=-3$, or $y=\frac{3}{2}$.

Case 1. If $\mu=0$. Then from (1) and (2) we have $\begin{cases} 2x = y+3 \\ 2y = x \end{cases}$.

So, $(x,y)=(2,1)$.

Case 2. If $\mu \neq 0$ and when $y = -3$ or $\frac{3}{2}$. Then from (5), $x = 0$ or $\pm\frac{3\sqrt{3}}{2}$.

Check that:

(x, y)	$f(x, y)$	$\mu = 1 - \frac{y+3}{2x}$ (from (1))	$\mu = 1 - \frac{x}{2y}$ (from (2))
$\left(\frac{3\sqrt{3}}{2}, \frac{3}{2}\right)$	-2.69	$\frac{2-\sqrt{3}}{2}$	$\frac{2-\sqrt{3}}{2}$
$\left(-\frac{3\sqrt{3}}{2}, \frac{3}{2}\right)$	20.69	$\frac{2+\sqrt{3}}{2}$	$\frac{2+\sqrt{3}}{2}$
$(0, -3)$	9	not applicable	1

Case 3. Obviously, we cannot have $y = -3$ and $y = \frac{3}{2}$.

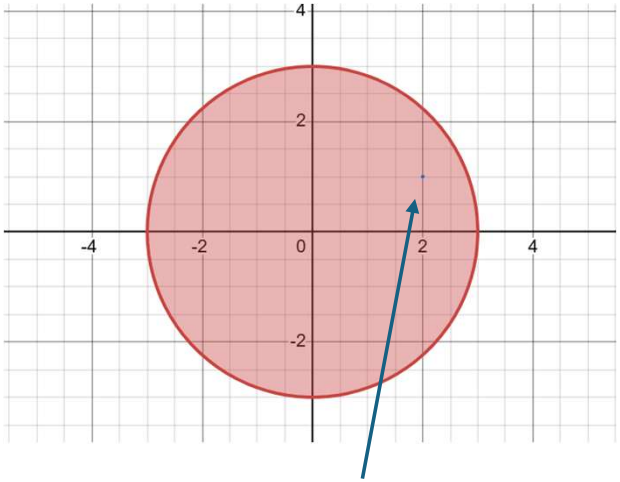
One can check these cases $(y, \mu) = (-3, 0)$ or $(\frac{3}{2}, 0)$.

From (2), we have $x = -6, 3$.

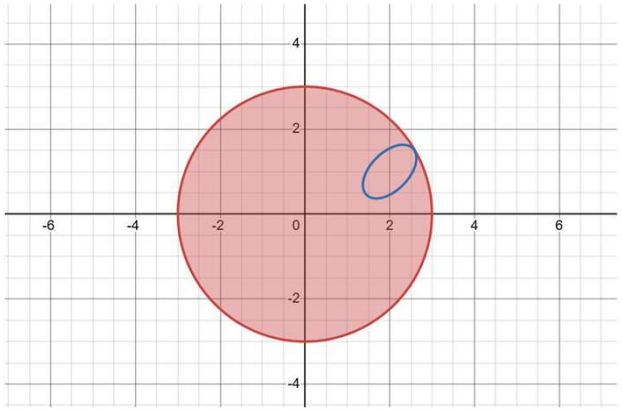
Check that:

(x, y)	$x^2 + y^2 \leq 9$ (3)	Conclusion
$\left(-6, 3\right)$	violates (3)	rejected
$\left(3, \frac{3}{2}\right)$	violates (3)	rejected

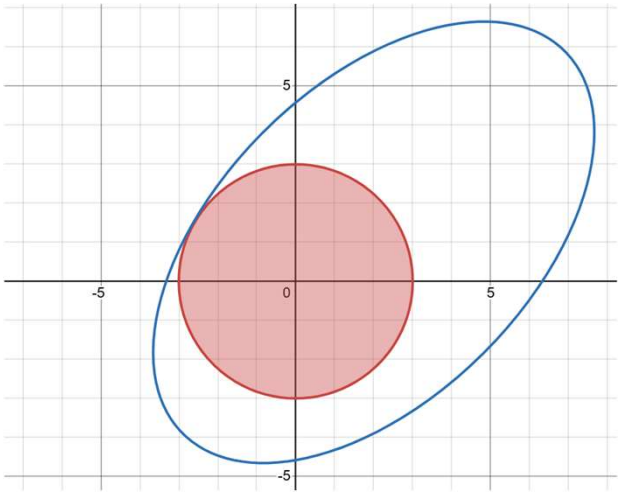
Conclusion. Hence, $f_{\max}\left(-\frac{3\sqrt{3}}{2}, \frac{3}{2}\right) = 20.69$.



$$f\left(2,1\right)=-3$$



$$f\left(\frac{3\sqrt{3}}{2},\frac{3}{2}\right)=9-\frac{27\sqrt{3}}{4}=-2.69$$



$$f\left(-\frac{3\sqrt{3}}{2},\frac{3}{2}\right)=9+\frac{27\sqrt{3}}{4}=20.69$$

Example 5

Minimise $(x-1)^2 + y^2$

Subject to $x \leq y^2$.

Solution. Let $f(x,y) = (x-1)^2 + y^2$ and $g(x,y) = x - y^2$ so that $g(x,y) \leq 0$.

$$\nabla\Big(f + \mu g\Big) = 0 \Rightarrow \begin{pmatrix} 2(x-1) + \mu \\ 2y - 2\mu y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$2(x-1) = -\mu \qquad (1)$$

$$(1-\mu)y = 0 \qquad (2)$$

$$\mu \geq 0 \qquad (3)$$

$$\mu(x - y^2) = 0 \qquad (4)$$

Take $(1) \times y = (2) : y = 0$ or $x = \frac{1}{2}$.

Case 1. $x = \frac{1}{2}$. Then $\mu = 1$ (from (1)) and $y = \pm \frac{1}{\sqrt{2}}$ (from (4)).

Check that $(x, y, \mu) = \left(\frac{1}{2}, \pm \frac{1}{\sqrt{2}}, 1\right)$ satisfies (1) to (4) and $g(x, y) \leq 0$.

Case 2. $y = 0$. Then either $x = 0$ or $\mu = 0$ (from (4)).

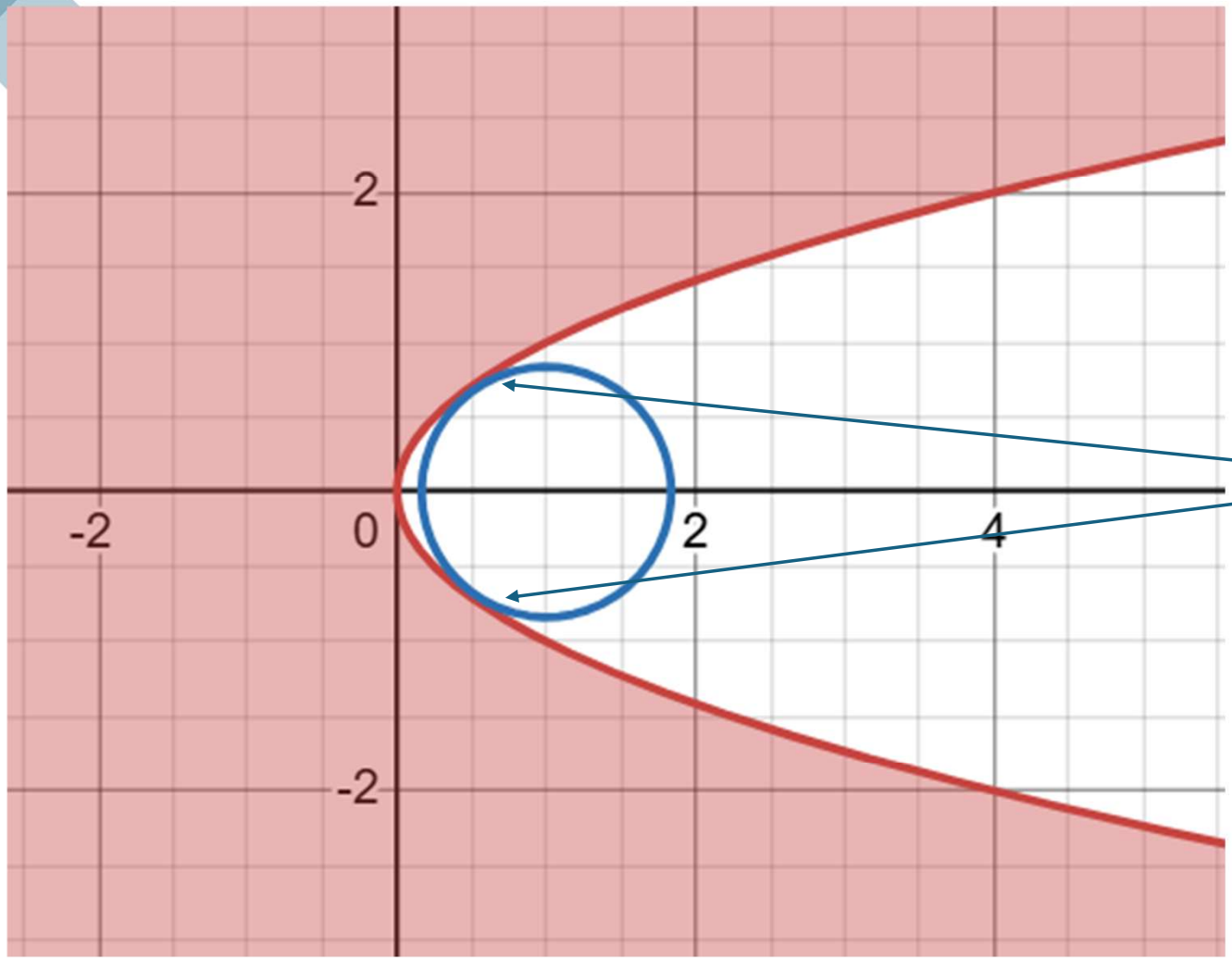
(a) If $x = 0$, then $\mu = 2$ (from (1)). So, $(x, y, \mu) = (0, 0, 2)$,

(b) If $\mu = 0$, then $x = 1$ (from (1)) but $g(1, 0) \not\leq 0$, a contradiction.

(c) If $x = \mu = 0$, then (1) is false, a contradiction.

Case 3. If $(x, y) = \left(\frac{1}{2}, 0\right)$. Then $g(\frac{1}{2}, 0) \not\leq 0$, a contradiction.

Conclusion. $f\left(\frac{1}{2}, \pm\frac{1}{\sqrt{2}}\right) = \frac{3}{4} < f(0, 0) = 1 \Rightarrow f_{\min} = \frac{3}{4}$ at $\left(\frac{1}{2}, \pm\frac{1}{\sqrt{2}}\right)$.



$$\left(\frac{1}{2}, \pm\frac{1}{\sqrt{2}}\right)$$

Example 6 (same as Example 5 but this is maximising)

Maximise $(x-1)^2 + y^2$

Subject to $x \leq y^2$.

Solution. Let $f(x,y) = (x-1)^2 + y^2$ and $g(x,y) = x - y^2$ so that $g(x,y) \leq 0$.

$$\nabla\big(f - \mu g\big) = 0 \Rightarrow \begin{pmatrix} 2(x-1) - \mu \\ 2y + 2\mu y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$2x - 2 = \mu \qquad (1)$$

$$-y = \mu y \qquad (2)$$

$$\mu \geq 0 \qquad (3)$$

$$\mu(x - y^2) = 0 \qquad (4)$$

Take $(1) \times y = (2) : y = 0$ or $x = \frac{1}{2}$.

Case 1. $x = \frac{1}{2}$. Then $\mu = 1$ (from (1)) and $y = \pm \frac{1}{\sqrt{2}}$ (from (4)).

Check that $(x, y, \mu) = \left(\frac{1}{2}, \pm \frac{1}{\sqrt{2}}, 1\right)$ satisfies (1) to (4) and $g(x, y) \leq 0$.

Case 2. $y = 0$. Then either $x = 0$ or $\mu = 0$ (from (4)).

(a) If $x = 0$, then $\mu = -2 \ngeq 0$ (from (1)) violates (3).

(b) If $\mu = 0$, then $x = 1$ (from (1)) but $g(1,0) \nleq 0$, a contradiction.

(c) If $x = \mu = 0$, then (1) is false, a contradiction.

Case 3. If $(x, y) = \left(\frac{1}{2}, 0\right)$. Then $g(\frac{1}{2}, 0) \nleq 0$, a contradiction.

Conclusion. KKT method failed to be applied.

Example 7

$$\begin{aligned} &\text{Minimise} && (x-1)^2 + y^2 \\ &\text{subject to} && g(x,y) = x^2 + y^2 + x + y = 0 \\ & && h(x,y) = x - y^2 \leq 0 \end{aligned}$$

Solution. *Necessary conditions*

$$-\nabla f = \lambda \nabla g + \mu \nabla h \Rightarrow \begin{cases} 2(x-1) = \lambda(2x+1) + \mu \\ -2y = \lambda(2y+1) - 2\mu y \end{cases}$$

$$\lambda(2x+1)+\mu = 2-2x \quad (1)$$

$$\lambda(2y+1)-2y\mu = -2y \quad (2)$$

$$x^2+y^2+x+y = 0 \quad (3)$$

$$x-y^2 \leq 0 \quad (4)$$

$$\mu \geq 0 \quad (5)$$

$$\mu(x-y^2) = 0 \quad (6)$$

From (6), we have two cases: $\mu = 0$ or $x = y^2$.

Case 1. $\mu = 0$. Note that $(x, y) = \left(-\frac{1}{2}, -\frac{1}{2}\right)$ is not a solution because it violates (3).

From (1) and (2) we have $\frac{2-2x}{2x+1} = \lambda = -\frac{2y}{2y+1} \Rightarrow \frac{1-x}{2x+1} = -\frac{y}{2y+1} \Rightarrow x - 3y = 1$ (7).

Take the intersection of (3), (4) and (7): $(x, y) = (-1.1708, -0.7236)$.

Substitute $(x, y) = (-1.1708, -0.7236)$ in (1) (or (2)): $(\lambda, \mu) = (-3.236, 0)$.

Case 2. $x = y^2$. From (3): $(x, y) = (0, 0)$ or $(0.2056, -0.4534)$.

Then from (1) and (2): $(x, y, \lambda, \mu) = (0, 0, 0, 2)$ or $(0.2056, -0.4534, 0.45, 0.9537)$.

Case 3. $x = y^2$ and $\mu = 0$. We see that $\mu \neq 0$ in Case 2. So this is impossible.

Check that

$$f(-1.1708, -0.7236) = 5.24$$

$$f(0, 0) = 5$$

$$f(0.2056, -0.4534) = 0.8366$$

